MCI WORKFORCE PLATFORM

Product Requirements Document

Unified Recruitment & Workforce Management System for On-Demand Workers

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TLDR SUMMARY

The MCI Workforce Platform is a unified, mobile-first web application designed to replace MCI Career Services' fragmented recruitment and workforce management workflow with a single end-to-end digital system.

The Problem

MCI currently operates across multiple disconnected tools — ZOHO CRM, Google Sheets, WhatsApp, and paid external job portals — to manage the full lifecycle of on-demand worker recruitment. This creates compounding inefficiencies: manual data transfer between platforms, overwhelming one-to-one WhatsApp communication for TAs, unreliable photo-based attendance, error-prone month-end payroll consolidation, and high external advertising costs. Workers have little reason to stay loyal to MCI when they are simultaneously exposed to competing recruiters on the same portals.

What the Platform Delivers

The solution covers three user groups — Workers, Talent Acquisition Managers (TAs), and Employer Clients — through purpose-built interfaces for each.

For workers, the platform provides a self-contained job marketplace where they can register, browse and apply for shifts matching their availability and location, receive automated reminders, clock in and out via GPS-verified photo capture, and track their earnings and payment status in real time. A recommendation engine surfaces relevant new jobs to keep workers engaged within MCI's ecosystem rather than drifting to external portals.

For TAs, the platform automates the most time-consuming parts of their workflow: the matching engine ranks eligible workers against job requirements (availability, proximity, skills, compliance, performance tier), and dispatches job notifications automatically. Real-time fill-rate dashboards replace Google Sheets; automated timesheet generation replaces manual consolidation; and one-click payroll exports replace error-prone manual bank transfers. Waitlist management and no-show auto-flagging handle last-minute disruptions without manual intervention.

For employer clients, a self-service portal allows direct job order submission and live recruitment tracking. Clients can flag preferred workers, monitor clock-ins in real time, endorse timesheets, and rate workers after shifts — all without needing to coordinate through MCI's BD team for every interaction.

Scope Boundaries

The MVP is scoped as a web-based application with mobile-responsive design, targeting a six-week build across three sprints. It includes core modules for worker onboarding, job posting and auto-matching, shift scheduling, GPS attendance verification, timesheet approval, payroll calculation, ZOHO CSV export, analytics dashboards, and a worker engagement tier system. Explicitly out of scope for the MVP are native mobile apps, live ZOHO API sync, WhatsApp Business API integration, direct banking API payouts, and Singpass MyInfo integration — all earmarked for Phase 2.

Success Criteria

The platform targets a 50% reduction in external recruitment costs and TA operational time, a 50% improvement in worker retention and repeat placement rates, near-zero payroll errors, and a 10% revenue uplift through improved client and candidate satisfaction — metrics directly aligned to the challenge's stated evaluation criteria.

1. Executive Summary

MCI Career Services Pte. Ltd. (MCI) is a leading Singapore-based recruitment agency specialising in on-demand, part-time, and ad-hoc workforce placement across logistics, events, retail, and F&B sectors. Despite strong market position, MCI's current operational workflow is deeply fragmented — relying on a patchwork of ZOHO CRM, Google Sheets, WhatsApp, and external job portals to manage recruitment, scheduling, attendance, and payroll. This creates massive inefficiencies, high operational costs, and weak candidate retention.

This document specifies the product requirements for the MCI Workforce Platform — a unified, mobile-first web application that digitises and automates the entire lifecycle from job posting to salary disbursement. The platform serves three primary user personas: Workers (gig/part-time candidates), Talent Acquisition Managers (TA), and Employer Clients. It consolidates all operations into a single system, replacing fragmented tools with one intelligent platform.

The MVP will be built as a web-based application designed with mobile-first principles and structured to seamlessly export data to ZOHO. This PRD defines the full scope of the MVP, the feature set, user stories, technical architecture, data models, and success metrics aligned to the IMDA Open Innovation Platform Challenge requirements.

Key Success Targets

Metric	Target
Reduction in external recruitment costs	**50% reduction**
Operational efficiency (recruitment, scheduling, payroll)	**50% time reduction**
Data entry and scheduling errors	**0% error rate (target)**
Worker retention and repeat placement	**50% improvement**
Internal staff satisfaction & retention	**Measurable increase beyond 3 months**
Revenue growth via improved satisfaction	**10% or more increase**

2. Problem Statement & Context

2.1 Business Context

Singapore's labour market has shifted significantly toward short-term, flexible staffing. Demand volatility from retail seasonal peaks, events, and logistics surges — combined with workers' preference for flexible income — has driven rapid growth in part-time, contract, and gig roles. Recruitment agencies like MCI play a critical role as intermediaries, matching supply and demand at scale and speed.

MCI currently manages large worker pools (segmented by skills, availability, and job type), processes job orders from employer clients on short notice, and handles end-to-end employment compliance, scheduling, and payroll. However, the operational model has not kept pace with the volume and complexity of demand.

2.2 Current Workflow Pain Points

The current workflow spans multiple disconnected tools and manual steps:

Pain Point	Description
Fragmented Systems	ZOHO CRM, Google Sheets, WhatsApp, and external portals (FastJobs, Jobstreet) are entirely disconnected. Data must be manually copied between platforms, causing errors and delays.
High Recruitment Costs	Job ads on multiple paid external portals are expensive and create dependency on third-party platforms.
Low Worker Retention	External portals expose workers to competing opportunities, making it difficult to build a loyal talent pool.
Communication Chaos	TAs manage hundreds of individual WhatsApp conversations with no centralised visibility, leading to missed messages and slow response times.
Manual Scheduling	Google Sheets-based master schedules require constant manual updates; any change in candidate availability creates a cascading update burden.
Unreliable Attendance	Photo-based attendance via WhatsApp is easily falsified; no GPS or geo-location verification.
Error-Prone Payroll	Month-end salary processing is fully manual, requiring data consolidation from multiple sources, verification, and bank transfers — prone to errors and delays.

2.3 Problem Statement

> *How might we develop an integrated recruitment and workforce management system that streamlines end-to-end operations, enhances internal efficiency, and strengthens candidate retention within MCI's ecosystem?*

3. Goals, Non-Goals & Constraints

3.1 Goals

- Deliver a unified web-based platform covering the full recruitment lifecycle: job posting, candidate matching, scheduling, attendance, timesheet approval, and payroll.
- Eliminate reliance on external job portals by creating an internal job marketplace for workers.
- Automate repetitive TA tasks: job broadcasting, candidate shortlisting, shift scheduling, timesheet generation, and payment processing.
- Implement GPS-verified, photo-timestamped attendance to eliminate falsification.
- Build worker engagement features (ratings, rewards, job recommendations) to drive retention.
- Provide employer clients with real-time visibility into staffing status and attendance.
- Export data to ZOHO CRM in CSV/structured formats for seamless backend integration.
- Deliver an accessible, multilingual (English & Mandarin) interface designed for users with varying digital literacy.
- Design architecture that is scalable to overseas clients and markets.

3.2 Non-Goals (MVP Scope Exclusions)

- Full native mobile app (iOS/Android) — the MVP will be mobile-responsive web; native app is a post-MVP consideration.
- Full direct ZOHO API integration — MVP will support CSV export/upload to ZOHO.
- Automated bank transfer/payroll disbursement integration with banking APIs — MVP will support one-click payment instructions generation; actual transfers remain manual.
- WhatsApp Business API integration — MVP will support in-app messaging; WhatsApp integration is an optional enhancement.
- Singpass MyInfo integration — planned as a Phase 2 feature.
- Multi-country payroll compliance — MVP targets Singapore labour law only.

3.3 Constraints

- Prototype must be demonstrable in Singapore by April 2026 submission deadline.
- Must be compatible with MCI's existing ZOHO CRM workflow formats.
- Must support users with low digital literacy — minimal learning curve, clear UI, and multilingual support.
- All data handling must comply with Singapore's Personal Data Protection Act (PDPA).

4. User Personas

The platform serves three distinct user types, each with different goals, pain points, and interaction patterns.

4.1 Persona 1: The Worker (Gig/Part-Time Candidate)

Attribute	Detail
Representative User	Lucy, 50-year-old ad-hoc F&B worker, limited digital literacy, Chinese-speaking
Primary Goals	Find and apply for nearby shifts quickly; receive timely payment; manage her schedule easily
Key Frustrations	Multiple platforms are confusing; language barriers; missed notifications; payment delays
Platform Needs	Simple onboarding, job browsing by location/availability, in-app shift calendar, clock-in/out via photo + GPS, real-time payment status, job recommendations
Device	Smartphone (primary), some desktop usage
Language	Chinese (Mandarin) or English

4.2 Persona 2: The Talent Acquisition Manager (TA)

Attribute	Detail
Representative User	Jared, TA Manager at MCI, manages multiple job orders and worker pools simultaneously
Primary Goals	Fill job orders quickly with qualified workers; manage attendance and timesheets; process payroll accurately
Key Frustrations	Platform-switching; manual scheduling; WhatsApp inbox overload; payroll errors; no real-time visibility
Platform Needs	Job order management dashboard, auto-matching engine, one-click approvals, real-time fill rate tracking, waitlist management, automated timesheet generation, ZOHO export
Device	Desktop/laptop (primary), mobile for urgent notifications

4.3 Persona 3: The Employer Client

Attribute	Detail
Representative User	Mr Wong, warehouse operations manager needing 20 packing staff for 2 weeks

Primary Goals	Submit job orders easily; track recruitment progress; verify attendance; endorse timesheets
Key Frustrations	No visibility into recruitment status; manual coordination with MCI BD team; unreliable attendance data
Platform Needs	Self-service job order submission, real-time fill tracking, candidate profile visibility, preferred worker flagging, shift notifications, attendance endorsement, timesheet approval
Device	Desktop/laptop (primary)

5. Feature Requirements

Features are categorised as Must Have (MVP), Should Have (MVP), and Could Have (Phase 2) using the MoSCoW method, mapped directly to the challenge requirements.

5.1 Worker Onboarding & Profile Management

5.1.1 Worker Registration & Profile

- Must Have: Email/phone-based self-registration with OTP verification.
- Must Have: Profile creation with fields: full name, NRIC/FIN (masked), contact details, bank account number, languages spoken, job type preferences, available working hours, skills/certifications.
- Must Have: Document upload for compliance: NRIC copy, work pass (for foreign workers), certifications, training records.
- Must Have: Profile completeness indicator to encourage workers to fill all fields.
- Should Have: Singpass MyInfo integration for one-tap profile population with verified data (Phase 2 — listed here for architecture planning).
- Must Have: Multilingual interface toggle (English / Simplified Chinese) available at registration and throughout the app.
- Must Have: Worker status field: Active, Inactive, Blacklisted (managed by TA).

5.1.2 Worker Segmentation & Pool Management (Admin/TA View)

- Must Have: Filterable, searchable worker database with segment tags: job type (F&B, logistics, admin, events), performance tier (standard, preferred, blacklisted), location, certifications.
- Must Have: Worker profile history: previous jobs, attendance record, ratings received, penalties.
- Must Have: Ability for TA to manually update worker status, add notes, or flag for follow-up.

5.2 Job Posting, Matching & Dispatch

5.2.1 Job Order Creation

- Must Have: Employer clients can submit job orders via a structured web form with fields: job title, job type, location, shift dates/times, number of workers required, hourly rate or per-shift pay, required skills/certifications, uniform/equipment requirements, special instructions.
- Must Have: TA can also create job orders on behalf of clients directly in the admin panel.
- Must Have: Support for multi-day and multi-shift job orders (e.g., 20 workers across 10 days, different shift timings per day).
- Must Have: Job order status tracking: Draft, Open, Partially Filled, Fully Filled, Completed, Cancelled.

5.2.2 Automated Matching Engine

- Must Have: Rule-based matching algorithm that scores and ranks eligible workers against a job order based on: availability (no conflicting shifts), location proximity (within

configurable radius in km), required skills/certifications, compliance status (valid documents), past performance rating, worker status (not blacklisted).

- Must Have: Configurable match weighting per job order type (e.g., proximity weighted higher for F&B, certifications weighted higher for events).
- Must Have: Support for multiple dispatch logic modes: First-Come-First-Served (worker who accepts first gets the slot); Performance-Based Priority (highest-rated workers notified first with a time window before broadcasting); Preferred Worker Priority (client-flagged preferred workers notified first).

5.2.3 Job Broadcasting & Worker Acceptance

- Must Have: Matched workers receive in-app push notifications and email alerts for available shifts matching their profile.
- Must Have: Workers can view full job details and accept/decline from the notification or the job marketplace.
- Must Have: Acceptance is automatically logged, and the worker is added to the confirmed shift roster.
- Must Have: Configurable acceptance window (e.g., 2-hour window before broadcasting more widely).
- Must Have: Workers who decline or do not respond within the window are de-prioritised for re-broadcast.
- Must Have: Waitlist functionality — additional applicants beyond the required headcount are placed in a ranked waitlist.
- Must Have: TA dashboard shows real-time fill rate per job order slot.

5.3 Scheduling & Shift Management

- Must Have: Auto-generated shift schedule for each job order, showing confirmed workers per slot.
- Must Have: TA can manually adjust shift assignments (add, remove, swap workers).
- Must Have: Workers receive shift confirmation notifications with full job details, location, and reporting instructions.
- Must Have: Shift reminder notifications: 24 hours before shift and 2 hours before shift.
- Must Have: Workers can confirm or cancel shift attendance via the app up to a configurable time threshold (e.g., 4 hours before shift).
- Must Have: Last-minute cancellations (within threshold) automatically trigger: flag on worker profile as a late cancellation, TA alert notification, auto-notification to next waitlisted worker.
- Must Have: Schedule view for workers: calendar view showing confirmed shifts, pending applications, and past shifts.
- Should Have: Conflict detection — system alerts TA if a worker is assigned to overlapping shifts.

5.4 Attendance Tracking & Verification

- Must Have: Workers clock in and clock out via the app using photo capture with automatic GPS coordinates and timestamp embedded.

- Must Have: GPS geo-fence check — system validates that the worker is within a configurable radius of the job site at the time of clock-in.
- Must Have: Geo-fence radius is configurable per job order by TA.
- Must Have: Clock-in photos and GPS coordinates are stored and viewable by TA and employer client.
- Must Have: If worker is outside the geo-fence on clock-in, system flags the record as 'Location Mismatch' and notifies TA.
- Must Have: Attendance status per worker per shift: Present, Absent (No-Show), Late, Unverified.
- Must Have: Automatic no-show detection: if worker does not clock in within a configurable grace period after shift start, system marks them as no-show and notifies TA.
- Must Have: Repeated no-shows trigger automated progressive warnings and, after configurable threshold (e.g., 3 no-shows), automatic blacklisting with TA notification.
- Must Have: Employer client receives real-time clock-in/clock-out notifications for their job site.

5.5 Timesheet Generation & Approval

- Must Have: System auto-generates timesheets per job order based on clock-in/clock-out records.
- Must Have: Timesheet shows: worker name, shift date, scheduled hours, actual hours worked, overtime (if applicable), hourly/shift rate, calculated gross pay.
- Must Have: Employer client reviews and endorses timesheets in the platform (approve all, approve individual, or flag a dispute).
- Must Have: TA reviews endorsed timesheets before releasing for payroll.
- Must Have: Dispute management: employer or TA can flag a timesheet entry for review with a comment; worker is notified.
- Must Have: Once fully approved, timesheet status changes to 'Verified' and becomes locked from editing.
- Must Have: Timesheets exportable as CSV and Excel for upload to ZOHO.

5.6 Payroll Processing & Payment

- Must Have: Automated salary calculation from verified timesheets, applying: hourly rate x actual hours, per-shift flat rate (for fixed-rate roles), overtime multiplier for hours beyond shift hours.
- Must Have: Multiple pay models supported: hourly rate, per-shift flat rate, mixed (hourly with per-shift minimum).
- Must Have: TA reviews the auto-calculated payroll summary before releasing payment instructions.
- Must Have: Payment instruction generation: system produces a structured file (CSV/Excel) with worker name, bank account, amount — for upload to the company's banking platform.
- Must Have: Workers receive in-app and email notification when payment is processed.
- Must Have: Workers can view their earnings history: per shift, per week, per month.

- Must Have: Payroll records are auditable — all changes are logged with timestamp and user ID.
- Should Have: Optional in-app e-wallet where salary is deposited for faster payout (Phase 2 consideration; architecture should allow for this).

5.7 Worker Engagement & Retention

- Must Have: In-app job marketplace for workers — browsable feed of available jobs matched to their profile, filterable by job type, location, date, and pay rate.
- Must Have: Recommendation engine — push similar or relevant job postings to workers based on their job history, preferences, and profile tags.
- Must Have: Performance rating system — after each shift, employer clients rate workers on a 1–5 scale with optional comments. Average rating is displayed on worker profiles.
- Must Have: Achievement and tier system — workers earn points per completed shift; higher tiers (Standard, Preferred, Elite) unlock priority job notifications and higher-visibility placements.
- Must Have: Transparent payment status dashboard for workers: Pending Verification, Employer Endorsed, TA Verified, Payment Processed, Paid.
- Must Have: In-app notifications centre — all shift updates, payment alerts, and new job recommendations in one place.
- Should Have: In-app two-way messaging between workers and TAs for shift-specific queries.

5.8 Employer Client Portal

- Must Have: Employer client self-service portal to submit and manage job orders.
- Must Have: Real-time recruitment dashboard per job order: number applied, shortlisted, confirmed, waitlisted, and slots remaining.
- Must Have: Access to candidate profiles for confirmed workers (name, profile photo, rating, past work history with the client).
- Must Have: Ability to flag preferred workers per job order — these workers are given first-priority notifications for that client's future jobs.
- Must Have: Attendance monitoring view: real-time clock-in/clock-out status during shift.
- Must Have: Timesheet review and endorsement workflow.
- Must Have: Post-shift worker rating and feedback submission.
- Must Have: Notification for: full staffing achieved, worker no-shows, clock-in alerts.

5.9 TA & Admin Dashboard

- Must Have: Unified job order management view with status filters.
- Must Have: Real-time fill rate tracker per job order.
- Must Have: Worker management panel: search, filter, view profiles, update status, manage blacklists.
- Must Have: Shift schedule view: calendar and list views for all active and upcoming shifts.
- Must Have: Attendance monitoring during live shifts with exception flagging.
- Must Have: Timesheet management: review, approve, flag disputes.

- Must Have: Payroll review and payment instruction generation.
- Must Have: ZOHO export: one-click CSV export of job orders, schedules, and timesheets.
- Must Have: Notification centre: all system alerts (no-shows, cancellations, disputes, blacklist triggers) in one place.
- Must Have: Audit log: view all system actions (who did what, when).

5.10 Analytics & Reporting

- Must Have: Dashboard KPIs: total active workers, active job orders, fill rate, current open positions, upcoming shifts.
- Must Have: Demand and supply charts: job orders vs. available workers over time.
- Must Have: Fill rate by job type, client, and time period.
- Must Have: Worker performance report: completion rate, average rating, no-show rate.
- Must Have: No-show and late cancellation trend analysis.
- Must Have: Payroll summary report: total hours, total pay disbursed, by period and by client.
- Should Have: Demand forecasting view — based on historical job order patterns, suggest likely manpower needs for upcoming periods.
- Should Have: Worker retention cohort analysis — track repeat placement rates over time.

5.11 Optional / Phase 2 Features

- Singpass MyInfo integration for fast, verified onboarding.
- WhatsApp Business API integration for last-minute shift alerts.
- In-app e-wallet for instant salary payouts.
- Native mobile app (iOS and Android).
- AI-powered no-show risk prediction (flag workers with high historical no-show probability before shift confirmation).
- Overseas client and multi-currency support.

6. User Stories & Acceptance Criteria

The following user stories define the functional expectations of the platform from each persona's perspective.

6.1 Worker User Stories

#	User Story	Acceptance Criteria
W-01	As a worker, I want to register with my phone number and create a profile so I can start applying for jobs.	OTP verification works; profile fields are saved correctly; worker receives welcome notification.
W-02	As a worker, I want to browse available jobs near me filtered by date and job type so I can find suitable shifts.	Job list loads with accurate distance and availability data; filters work correctly; job details are complete.
W-03	As a worker, I want to apply for a shift and be notified of my application status so I know if I'm confirmed.	Application submitted successfully; status updates (Pending, Confirmed, Waitlisted, Rejected) delivered via push and email.
W-04	As a worker, I want to receive shift reminders 24 hours and 2 hours before my shift so I don't forget.	Notifications sent at correct times; include job location, reporting instructions, and contact.
W-05	As a worker, I want to clock in by taking a photo at the job site so my attendance is recorded accurately.	Photo captured with GPS and timestamp; geo-fence check runs; system records clock-in; employer notified.
W-06	As a worker, I want to see my earnings status update in real time so I know when I'll be paid.	Dashboard shows per-shift status (Pending, Endorsed, Verified, Paid); payment notification sent when processed.
W-07	As a worker, I want to see personalised job recommendations so I can find relevant shifts without searching.	Recommendations generated based on job history and profile; refreshed after each completed shift.
W-08	As a worker who speaks Chinese, I want to use the app in Mandarin so I can navigate it confidently.	Full interface translates to Simplified Chinese; all notifications sent in preferred language.

6.2 Talent Acquisition Manager User Stories

#	User Story	Acceptance Criteria
T-01	As a TA, I want to create and publish a job order so I can start receiving worker applications.	Job order created with all required fields; published status broadcasts to matched workers; order visible in dashboard.
T-02	As a TA, I want the system to auto-match and rank workers for a job so I can approve quickly.	Ranked candidate list generated based on availability, proximity, rating, and compliance; TA can approve in one click.
T-03	As a TA, I want to see real-time fill rate for all open jobs so I can prioritise action.	Dashboard shows fill rate per job order (e.g., 15/20 filled) updated in real time.

T-04	As a TA, I want to be immediately alerted when a confirmed worker cancels so I can find a replacement fast.	Push notification sent to TA within 1 minute of cancellation; waitlisted worker auto-notified; slot re-opened.
T-05	As a TA, I want to review and approve verified timesheets before releasing payroll so I maintain oversight.	Timesheet list shows status; TA can approve individually or in bulk; approved records lock and trigger payroll calculation.
T-06	As a TA, I want to export timesheets and payroll data to CSV so I can upload to ZOHO.	Export generates correctly formatted CSV; includes all required ZOHO fields; download completes within 10 seconds.
T-07	As a TA, I want to blacklist a worker after repeated no-shows so unreliable workers are removed from the pool.	Blacklist action logged; worker status updated; worker no longer receives job notifications; TA notified of auto-blacklists.

6.3 Employer Client User Stories

#	User Story	Acceptance Criteria
E-01	As a client, I want to submit a job order online so I don't have to call or email MCI every time.	Job order form submittable online; TA receives notification; order visible in client dashboard immediately.
E-02	As a client, I want to track how many workers have been confirmed for my job in real time.	Client dashboard shows confirmed count vs. required; updated live as workers accept.
E-03	As a client, I want to flag preferred workers so they are prioritised for my future jobs.	Preferred flag saved per worker per client; preferred workers notified first for that client's subsequent job orders.
E-04	As a client, I want to see who has clocked in on my site in real time so I can manage operations.	Live attendance view shows each worker's clock-in status, time, and photo; refreshes automatically.
E-05	As a client, I want to endorse timesheets after shifts so that workers can be paid accurately.	Timesheet endorsement flow is accessible; client can approve/dispute individual entries; TA notified on completion.
E-06	As a client, I want to rate workers after shifts so MCI can track performance.	Rating widget available post-shift; 1-5 stars with optional comment; rating saved to worker profile immediately.

7. Technical Architecture

7.1 Architecture Overview

The MCI Workforce Platform is designed as a mobile-first, responsive web application using a modern full-stack architecture. The system is built for rapid MVP delivery while maintaining a clear path to scale — supporting increased job volumes, additional markets, and native mobile app packaging.

Recommended Stack

Layer	Technology Choice
Frontend	React.js (TypeScript) — mobile-responsive, PWA-capable; Tailwind CSS for fast, consistent UI
Backend	Python Fast API — REST API with clear endpoint structure
Database	PostgreSQL — relational model ideal for recruitment data (workers, jobs, shifts, timesheets); Redis for session management and real-time notifications
Authentication	JWT-based auth with role-based access control (Worker, TA, Client, Admin); OTP via SMS (Twilio or AWS SNS)
File Storage	AWS S3 or equivalent — for worker documents (NRIC, certifications) and attendance photos
Notifications	Firebase Cloud Messaging (FCM) for in-app push; SendGrid for email notifications
Geolocation	Google Maps API / Mapbox — for geo-fence calculations, proximity matching, and location display
Hosting	AWS — containerised with Docker; CI/CD pipeline via GitHub Actions
Export Integration	CSV/Excel export (Papa Parse / SheetJS) formatted to ZOHO import schemas

7.2 System Architecture Diagram

The architecture follows a three-tier model:

- Presentation Layer: React PWA accessible from desktop and mobile browsers. Each user role has a distinct layout and navigation path (Worker App, TA/Admin Dashboard, Client Portal).
- Application Layer: REST API server handling business logic, authentication, matching engine, notification dispatch, and export generation. Modular service design allows individual services (matching, payroll, notifications) to be scaled independently.
- Data Layer: PostgreSQL database for structured data; S3 for file storage; Redis for caching and real-time event queuing.

The application is containerised using Docker, enabling consistent deployment across staging and production environments. All services communicate via internal APIs, with external integrations (SMS, email, geo-location) handled via third-party SDKs with retry logic.

7.3 Key Data Models

Worker

- id, name, email, phone, nric_masked, bank_account, languages[], job_type_preferences[], skills[], certifications[], location_lat, location_lng, rating_avg, tier (standard/preferred/elite), status (active/inactive/blacklisted), documents[], created_at, updated_at

Job Order

- id, client_id, created_by (TA user_id), title, job_type, location_address, location_lat, location_lng, shift_dates[], workers_required, hourly_rate, per_shift_rate, pay_model (hourly/per_shift/mixed), required_skills[], required_certifications[], geofence_radius_m, dispatch_mode (fcfs/performance/preferred), status, created_at, updated_at

Shift

- id, job_order_id, date, start_time, end_time, worker_id, status (pending/confirmed/completed/cancelled/no_show), confirmed_at, cancellation_reason, waitlisted (bool)

Attendance Record

- id, shift_id, worker_id, clock_in_time, clock_in_photo_url, clock_in_lat, clock_in_lng, clock_in_geofence_valid (bool), clock_out_time, clock_out_photo_url, clock_out_lat, clock_out_lng, total_hours, status (unverified/present/absent/flagged)

Timesheet Entry

- id, job_order_id, shift_id, worker_id, attendance_record_id, scheduled_hours, actual_hours, overtime_hours, hourly_rate, gross_pay, status (pending/employer_endorsed/ta_verified/payment_released), dispute_flag (bool), dispute_note, employer_endorsed_at, ta_verified_at

Payment Record

- id, worker_id, timesheet_entry_ids[], total_amount, payment_status (pending/processing/paid), payment_instructions_file_url, paid_at, created_at

7.4 Security & Compliance

- All PII (NRIC, bank account) stored encrypted at rest using AES-256.
- HTTPS enforced across all endpoints; HSTS headers enabled.
- Role-based access control (RBAC) — workers cannot access TA or client data; clients cannot access other clients' data.
- Attendance photos stored in private S3 buckets with signed URLs (expiry: 24 hours for display).
- PDPA compliance: data retention policy, right to request deletion, consent captured at registration.
- Audit log records all state changes with user ID, timestamp, and previous/new values.
- Input validation and rate limiting on all API endpoints to prevent abuse.

7.5 ZOHO Integration Design

The MVP delivers ZOHO integration via structured CSV export rather than live API sync (live sync is Phase 2). The export module generates files that match ZOHO's import schema for:

- Job Orders — mapped to ZOHO Deals/Projects fields.
- Candidate Timesheets — mapped to ZOHO People/HR timesheet module fields.
- Payroll Records — mapped to ZOHO Payroll/Expense fields.

TA can trigger export from the dashboard at any time. The Phase 2 roadmap includes a ZOHO API connector for real-time bidirectional sync.

7.6 Scalability Considerations

- Database: Read replicas for high-traffic reporting queries; connection pooling via PgBouncer.
- Matching engine: Job queue (e.g., BullMQ with Redis) allows async matching to scale independently of web server.
- Notifications: FCM and email are asynchronous and queue-based — high volume does not block user-facing operations.
- Multi-region: Cloud hosting setup allows geographic expansion by deploying additional regional instances with data residency compliance.
- Multi-currency and multi-language: i18n framework (i18next) and currency abstraction layer allow expansion without core code changes.

8. UI/UX Requirements

8.1 Design Principles

- Mobile-First: All interfaces designed for 375px viewport first, then scaled to desktop. Worker-facing features must be fully functional on small screens.
- Accessibility: WCAG 2.1 AA compliance target. High contrast ratios, large tap targets (minimum 44x44px), clear labels.
- Low Digital Literacy Support: Minimal text, icon-assisted navigation, step-by-step onboarding flows with progress indicators, contextual help tooltips.
- Multilingual: Full interface in English and Simplified Chinese. Language selector persistent across sessions. All error messages, notifications, and system text translated.
- Consistent Design System: Component library with standardised buttons, forms, cards, and modals to ensure consistency and rapid development.

8.2 Worker App Key Screens

- Onboarding: Registration form (step-by-step, one field per step on mobile), document upload, language selector, profile completion checklist.
- Home / Dashboard: Available shifts count, next upcoming shift, earnings summary, recent notifications.
- Job Marketplace: Card-based job listing with job type icon, pay rate, distance, and shift date/time. Filter bar for job type, date range, location radius, pay range.
- Job Detail: Full description, map view of job location, apply button, estimated earnings calculator.
- My Shifts: Calendar view and list view. Upcoming (confirmed), Applied (pending), Completed, Cancelled tabs.
- Clock In/Out: Large, prominent button. Camera capture flow with geo-validation status indicator. Confirmation screen post clock-in.
- Earnings: Per-shift breakdown, payment status badges, total month earnings, export to PDF option.
- Profile: Edit details, upload/update documents, language preference, notification settings.

8.3 TA Dashboard Key Screens

- Overview: Summary KPIs (open jobs, fill rate, today's active shifts, pending timesheets, pending payments), alerts panel.
- Job Orders: Table view with status filters, create new order button, inline fill rate indicator per row.
- Job Order Detail: Candidate list with match score, quick approve/reject actions, waitlist management, shift schedule grid.
- Attendance Monitor: Live view during active shifts — worker list, clock-in status, location validation flag, photo thumbnails.
- Timesheets: Filterable list with status and dispute flags; bulk approve function; dispute resolution panel.

- Payroll: Summary table per worker; review and release; download payment instructions CSV.
- Workers: Searchable/filterable database; individual profile view with history; status management.
- Analytics: Chart dashboard with configurable date range; exportable report.

8.4 Client Portal Key Screens

- Job Orders: List of submitted orders with status; create new order button.
- Job Order Detail: Recruitment progress, confirmed worker roster, preferred worker flags.
- Live Shift: Real-time clock-in status grid during active shifts.
- Timesheets: Review and endorse view, dispute flag option.
- Worker Ratings: Post-shift rating submission for each worker.

9. Matching Algorithm Specification

9.1 Overview

The matching engine is a rule-based scoring algorithm that evaluates all eligible workers against a job order's requirements and produces a ranked candidate list for TA review (or direct auto-dispatch, depending on dispatch mode configuration).

9.2 Eligibility Filter (Hard Rules — Must Pass All)

- Worker status is Active (not inactive or blacklisted).
- Worker has no confirmed shift overlapping with the requested shift time.
- Worker has all required certifications and valid compliance documents.
- Worker has confirmed availability for the requested shift date/time (availability is set in profile).

9.3 Scoring Model (Soft Ranking)

Factor	Default Weight	Configurable?
Location Proximity (closer = higher score)	30%	Yes — per job type
Past Performance Rating (higher rating = higher score)	25%	Yes
Job Type Match (matches preferences = bonus)	20%	Yes
Shift Completion Rate (history of completing shifts)	15%	Yes
Worker Tier (Elite > Preferred > Standard)	10%	Yes

TAs can override weights per job order type in the admin configuration panel. All weights must sum to 100%.

9.4 Dispatch Modes

- First-Come-First-Served (FCFS): All eligible workers above a minimum score threshold are notified simultaneously. Shift slots are filled as workers accept, in order of acceptance time.
- Performance-Based Priority: Top N workers (configurable, e.g., top 20%) are notified first with an exclusive acceptance window (e.g., 2 hours). Remaining eligible workers are notified if slots remain unfilled after the window.
- Preferred Worker Priority: Workers flagged as preferred by the employer client are notified first. After a priority window, the system falls back to performance-based or FCFS for remaining slots.

9.5 Recommendation Engine

Worker job recommendations are generated using a lightweight collaborative filtering and profile-matching approach:

- Jobs that match the worker's preferred job types and past job history are given higher recommendation scores.
- Jobs with a high fill urgency (fewer slots remaining relative to time before shift) are surfaced more prominently.
- Jobs the worker has previously applied to or completed at the same client are surfaced as 'Return Opportunity' recommendations.
- Recommendations are refreshed daily and after each completed shift. Workers receive a push notification when new relevant jobs are available.

10. Integrations

Integration	Purpose	Phase
ZOHO CRM	CSV export of job orders, timesheets, payroll data in ZOHO-compatible format	**MVP**
Google Maps / Mapbox	Geo-fence validation, distance calculation for matching, job location display	**MVP**
Firebase Cloud Messaging	In-app push notifications for job broadcasts, shift reminders, no-show alerts	**MVP**
SendGrid / AWS SES	Transactional email notifications (registration, shift confirmation, payment)	**MVP**
Twilio / AWS SNS	OTP SMS for registration and login verification	**MVP**
AWS S3	Secure storage for attendance photos and worker compliance documents	**MVP**
Singpass MyInfo	Verified worker profile population via government identity system	**Phase 2**
WhatsApp Business API	Last-minute shift alerts and TA-worker messaging for urgent updates	**Phase 2**
Banking API (PayNow/FAST)	Direct salary disbursement to worker bank accounts or PayNow IDs	**Phase 2**

11. Success Metrics & KPIs

The following metrics directly map to the challenge's stated Metrics of Success and will be measured at regular intervals post-launch.

KPI	Baseline (Current)	Target	Measurement Method
External job portal spend	MCI's current spend	-50%	Finance report comparison
TA time on scheduling/payroll	Estimated hours/week	-50%	TA time tracking survey
Data entry errors (scheduling, payroll)	Current error rate	0% (target)	System audit log
Worker repeat placement rate	Current rate	+50%	Platform placement history
Job order fill time (avg hours to fill)	Current avg	-30%	Job order timestamps
Worker app session frequency	N/A (new metric)	Weekly active users target	App analytics
Timesheet dispute rate	Current manual dispute rate	<5%	Dispute flag count / total timesheets
Staff (TA) satisfaction score	Current score	Measurable improvement	Quarterly survey
Revenue growth	Baseline revenue	+10%	Finance reporting

12. Development Roadmap

12.1 MVP Phased Delivery Plan

The MVP is planned across three sprints targeted to deliver a demonstrable prototype by mid-April 2026.

Sprint	Theme	Deliverables	Weeks
Sprint 1	Foundation & Core Auth	Project setup, DB schema, auth system (worker/TA/client roles), worker registration & profile, multilingual toggle (EN/ZH), job order creation (TA)	1–2
Sprint 2	Matching, Dispatch & Attendance	Matching engine, job marketplace (worker), push notifications, shift scheduling, GPS attendance (clock-in/out with photo & geo-fence), waitlist logic, no-show auto-flagging, shift reminders	3–4
Sprint 3	Payroll, Analytics & Polish	Auto-timesheet generation, timesheet approval workflow, payroll calculation, payment instructions export, ZOHO CSV export, analytics dashboard, worker ratings & tier system, job recommendations, client portal, UI polish, demo data seeding, UAT	5–6

Target demo-ready date: 10 April 2026 (1 week before submission deadline for buffer). Submission: 17 April 2026.

12.2 Post-MVP Roadmap (Phase 2)

- Native iOS and Android app packaging (React Native or Capacitor wrapper).
- Live ZOHO API bidirectional sync.
- Singpass MyInfo integration for instant verified onboarding.
- WhatsApp Business API for last-minute alerts.
- In-app e-wallet and direct PayNow disbursement.
- AI-powered no-show risk prediction model.
- Multi-language expansion (Malay, Tamil).
- Overseas market expansion with multi-currency payroll support.

13. Risks & Mitigations

Risk	Mitigation	Severity
Scope creep beyond 6-week timeline	Strict MoSCoW prioritisation; Phase 2 backlog well-defined; weekly scope reviews.	High
GPS geo-fence accuracy in dense urban environments	Use configurable radius with TA override for flagged anomalies; default 200m radius with buffer.	Medium
Worker adoption of new platform (low digital literacy)	Guided onboarding, multilingual UI, tutorial videos, simplified UX design, TA-assisted registration option.	Medium
PDPA compliance for sensitive worker data (NRIC, bank)	Encrypt PII at rest; PDPA consent flow at registration; data retention policy; privacy policy document.	High
ZOHO export format mismatch	Request ZOHO import template from MCI early; test export compatibility in Sprint 2.	Medium
Third-party API reliability (Geo, SMS, notifications)	Implement retry logic and fallback (e.g., email fallback if SMS fails); monitor with uptime alerting.	Low
Workers gaming GPS (using VPN or fake location)	Photo verification combined with GPS; employer can flag anomalies; TA review of flagged records.	Low

14. Open Questions & Clarifications Needed

- What is the exact ZOHO CRM module structure and field naming used by MCI for job orders, timesheets, and payroll? A ZOHO export template would allow export mapping to be built accurately.
- What is the current volume of active workers in MCI's pool and the approximate monthly job order count? This informs database sizing and matching engine performance requirements.
- Are there specific certifications or compliance documents that vary by job type? Full list needed to configure the compliance requirement matrix.
- What is MCI's preferred bank transfer method for payroll? (GIRO batch file, PayNow, internet banking CSV) — this determines the payment instruction file format.
- Does MCI want the employer client portal to be accessible to all clients, or only specific accounts? Client access control model needs to be agreed.
- Are there any existing branding guidelines (logo, colour palette) to be applied to the platform UI?
- What is the acceptable geo-fence radius default for different job types (indoor venues vs. outdoor logistics sites)?

Appendix: Glossary

Term	Definition
BD	Business Development — MCI's client-facing team that onboards employers and manages client relationships.
TA	Talent Acquisition Manager — MCI's internal recruiter responsible for sourcing, matching, and managing worker pools.
Geo-fence	A virtual geographic boundary around a job site location. Used to validate worker clock-in proximity.
FCFS	First-Come-First-Served — dispatch mode where job slots are filled in order of worker acceptance.
ZOHO	MCI's existing CRM platform used for tracking job orders, timesheets, and payroll records.
PWA	Progressive Web App — a web application designed to behave like a native mobile app when accessed from a mobile browser.
OTP	One-Time Password — a security code sent via SMS used for login and registration verification.
PDPA	Personal Data Protection Act — Singapore's primary data protection legislation.
Singpass	Singapore's national digital identity platform, used for verified identity authentication.
MoSCoW	Prioritisation framework: Must Have, Should Have, Could Have, Won't Have.
RBAC	Role-Based Access Control — security model where system access is determined by user role.
KPI	Key Performance Indicator — measurable metric used to evaluate success.